Exercise 5 :

a)

. Let $e_t = E[S_t] = E[\mu + S_{t-1} + X_t]$

$= \mu + E[S_{t-1}] + \underbrace{E[X_t]}_{0}$ by linearity

$e_t = \mu + e_{t-1} = \mu + (\mu + e_{t-2})$ and so on

(e_t) is an arithmetic sequence then its general form is $e_t = t\mu$ (trivial recurrence)

so

$$\boxed{E[S_t] = t\mu}$$

. Let's first write (S_t) in its general form.

$S_t = \mu + (\mu + S_{t-2} + X_{t-1}) + X_t$

$= 2\mu + X_t + X_{t-1} + S_{t-2}$ and so on

by trivial recurrence we can show that

$$S_t = t\mu + \sum_{i=1}^{t} X_i$$